AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Project Report

Assignment Title:	Python Project Report (Mid)		
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Course Title:	Programming in Python		
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Semester:	Spring 2025-26	Course Teacher:	**MD. TANZEEM RAHAT**

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	Total Marks	

PROJECT REPORT

Personal Expense Tracker

Desktop GUI Application

Field	Details
Project Name	Personal Expense Tracker
Track	Desktop App
Language	Python 3.1
GUI Framework	Tkinter
Storage	JSON
Architecture	Object-Oriented Programming (OOP)
Submission Date	March 28, 2026

1. Setup Instructions

Follow the steps below to set up and run the application on local machine.

Step-by-Step Setup

1. **Create virtual environment** — python -m venv .venv
2. **Install dependencies** — pip install -r requirements.txt
3. **Run the application** — python ui.py

Command Reference

```
# 1. Create & activate virtual environment
python -m venv .venv

# Windows
.venv\Scripts\activate

# macOS / Linux
source .venv/bin/activate

# 2. Install dependencies
pip install -r requirements.txt

# 3. Launch the app
python ui.py
```

2. Features

The Personal Expense Tracker provides a comprehensive set of features for managing daily finances through an intuitive graphical interface.

Feature	Description
CRUD Operations	Add, Edit, and Delete expenses with full data management support
Category Filtering	Filter expenses by predefined or custom categories
Date Filtering	Narrow down expenses by specific date ranges
Sorting	Sort by date, amount, title, or category (ascending/descending)
Summary Report	View total spending, average expense, and top categories at a glance

3. OOP Design

The application is architected using core Object-Oriented Programming principles, promoting clean separation of concerns, reusability, and maintainability.

Classes Used

Class	Role
Expense	Entity — Represents a single expense record with all its attributes
JSONStorage	Storage Layer — Handles reading and writing data to the JSON file
ExpenseManager	Service / Manager — Business logic: CRUD, search, sort, filter, report
ExpenseTrackerApp	UI Controller — Manages the Tkinter GUI and user interactions

OOP Concepts Applied

- Encapsulation — Each class hides its internal data and exposes only necessary interfaces
- Composition — ExpenseTrackerApp composes ExpenseManager; ExpenseManager composes JSONStorage
- Modular Design — Each class has a single, well-defined responsibility (SRP)
- Abstraction — UI layer is decoupled from storage and business logic

4. Functionality

The system allows users to manage their personal expenses efficiently through a clean graphical interface. All core operations are fully implemented and integrated.

Operation	Implementation Status
Create Expense	✓ Fully implemented with form validation
Read / View Expenses	✓ Displayed in sortable, filterable table
Update Expense	✓ Edit dialog pre-populated with existing data
Delete Expense	✓ Confirmation prompt before permanent removal
Reporting	✓ Total, average, and top category breakdown

5. Data Persistence

All expense data is persisted between sessions using a lightweight JSON-based storage system — requiring no external database or server.

Aspect	Detail
Storage Format	JSON (JavaScript Object Notation)

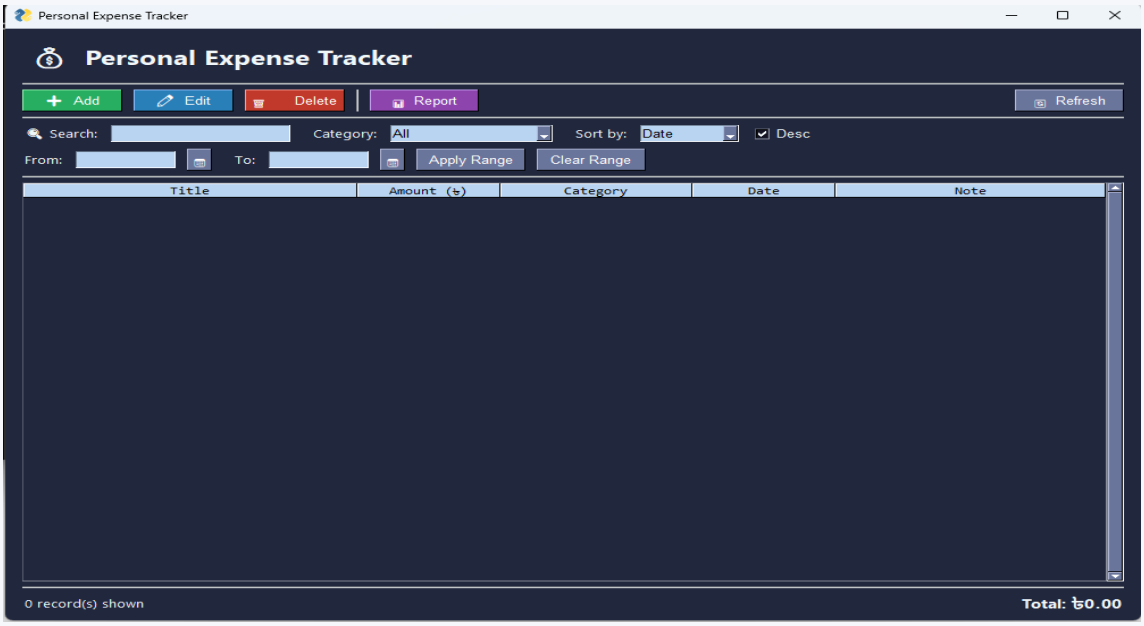
File Location	data/expenses.json
Load Strategy	Automatically loaded from disk when the application starts
Save Strategy	Automatically saved to disk after every create, update, or delete operation
Advantages	Human-readable, portable, zero configuration required

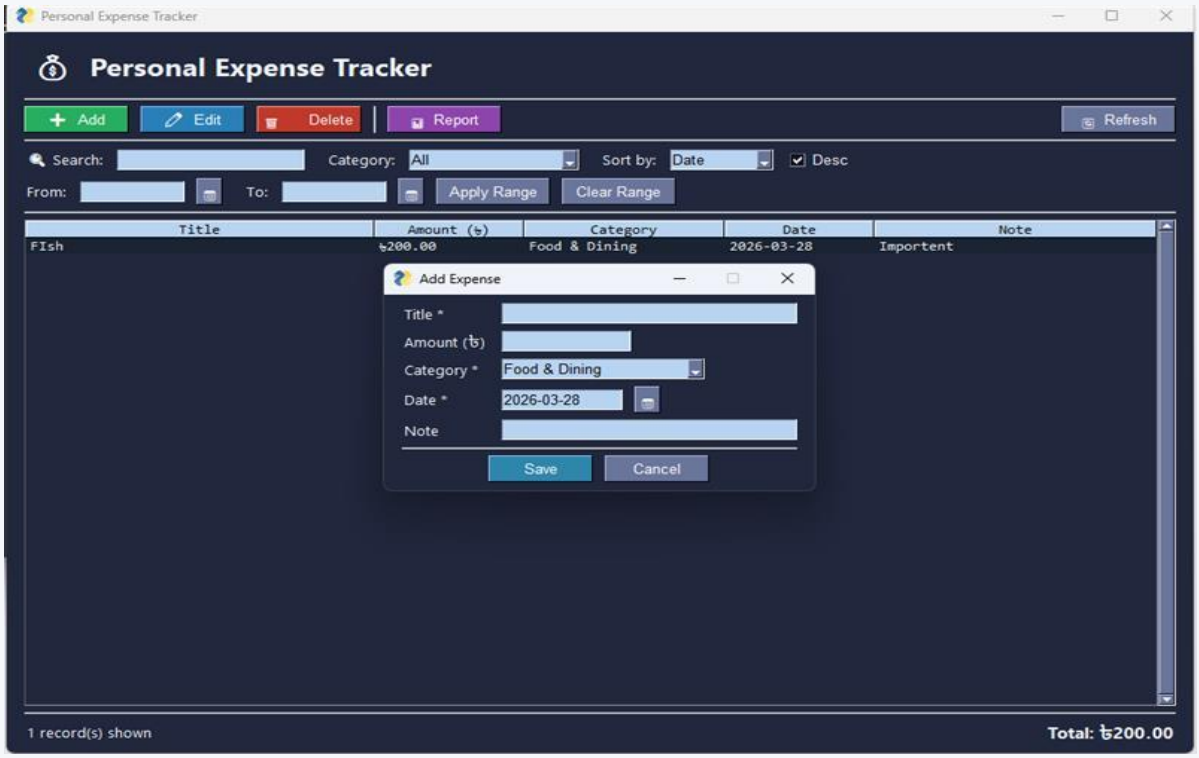
6. Validation & Error Handling

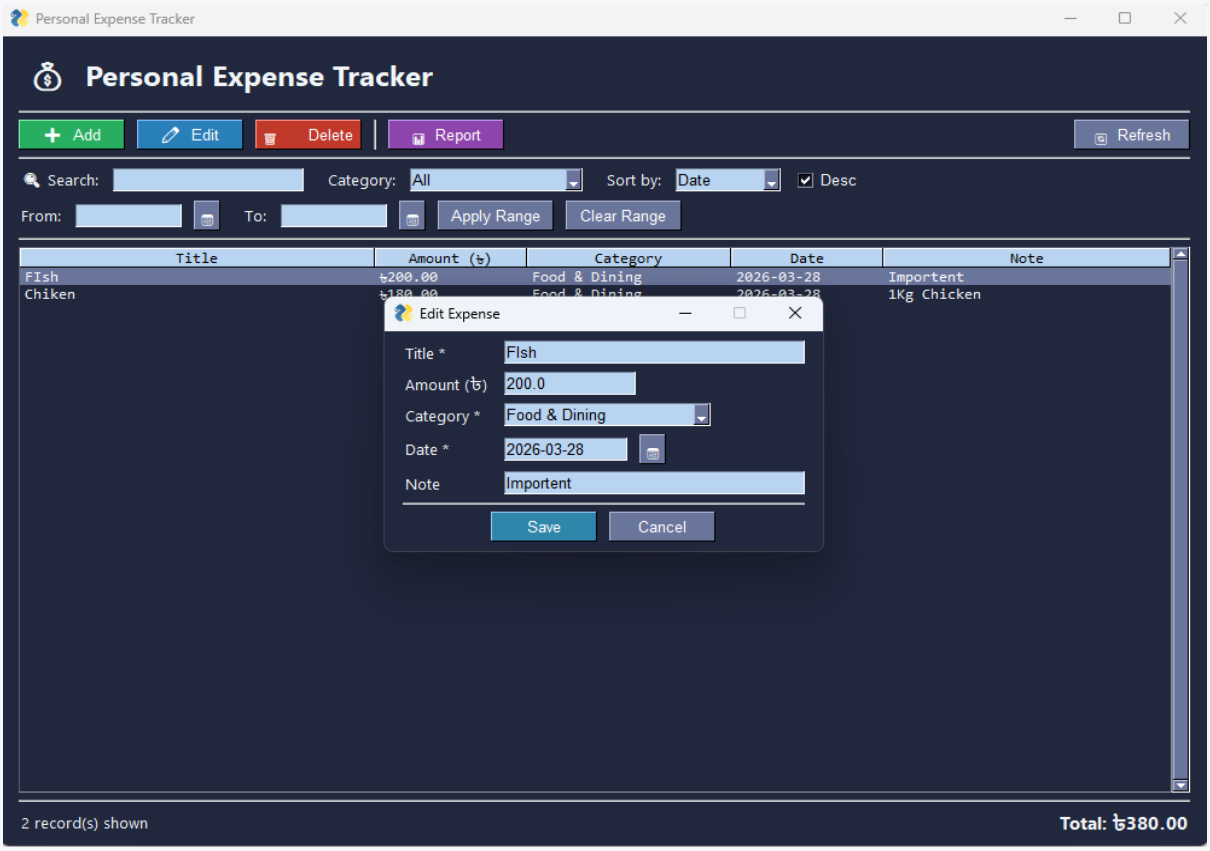
Robust input validation and graceful error handling ensure a stable and user-friendly experience, preventing data corruption and unexpected crashes.

- Empty Input Prevention — Required fields are validated before any data is saved
- Number Validation — Amount fields enforce numeric input only; invalid entries prompt a clear error message
- Crash Prevention — All critical operations are wrapped in try-except blocks
- User-Friendly Messages — Descriptive dialogs guide users when errors occur
- Graceful Degradation — Missing or corrupted data files are handled without application failure

7. Screenshots




Personal Expense Tracker
Personal Expense Tracker
Add
Edit
Delete
Report
Refresh
Search:
Category:
All
Sort by:
Date
Desc
From:
To:
Apply Range
Clear Range
Title
Amount (৳)
Category
Date
Note
FIsh
৳200.00
Food & Dining
2026-03-28
Importent
Add Expense
Title *
Amount (৳)
Category *
Food & Dining
Date *
2026-03-28
Note
Save
Cancel
1 record(s) shown
Total: ৳200.00


Personal Expense Tracker
Personal Expense Tracker
Add
Edit
Delete
Report
Refresh
Search:
Category:
All
Sort by:
Date
Desc
From:
To:
Apply Range
Clear Range
Title
Amount (৳)
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Date
Note
FIsh
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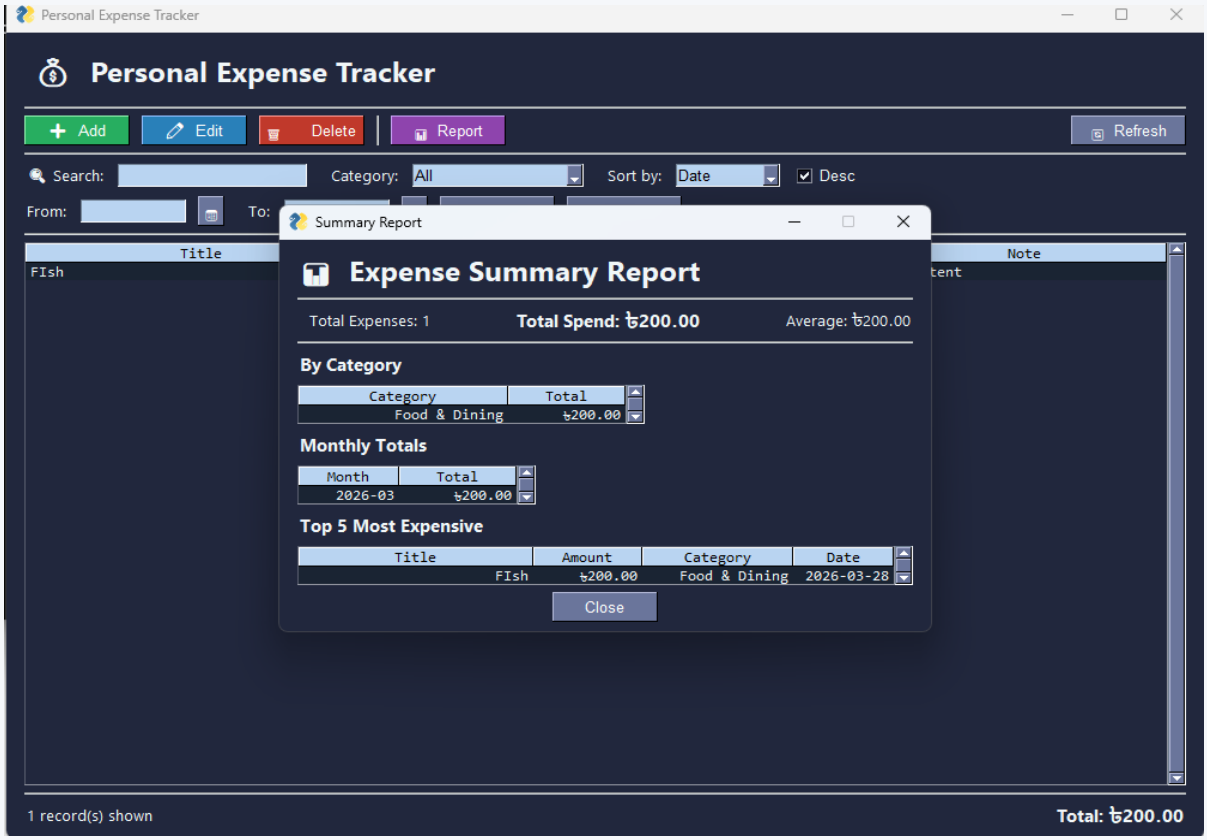
Food & Dining
2026-03-28
Importent
Chiken
1Kg Chicken
Edit Expense
Title *
FIsh
Amount (৳)
200.0
Category *
Food & Dining
Date *
2026-03-28
Note
Importent
Save
Cancel
2 record(s) shown
Total: ৳380.00



8. Conclusion

Key strengths of this project:

- Clean OOP architecture with clearly defined class responsibilities
- Responsive and intuitive GUI built entirely with Python's standard library
- Zero-dependency persistence via JSON — no external database required
- Comprehensive validation and error handling for a production-quality feel
- Full CRUD functionality supplemented by search, sort, filter, and reporting

— End of Report —